

Ex.no:6	DATA VISUALIZATION USING POWER BI (HOTEL BOOKING)

Aim:

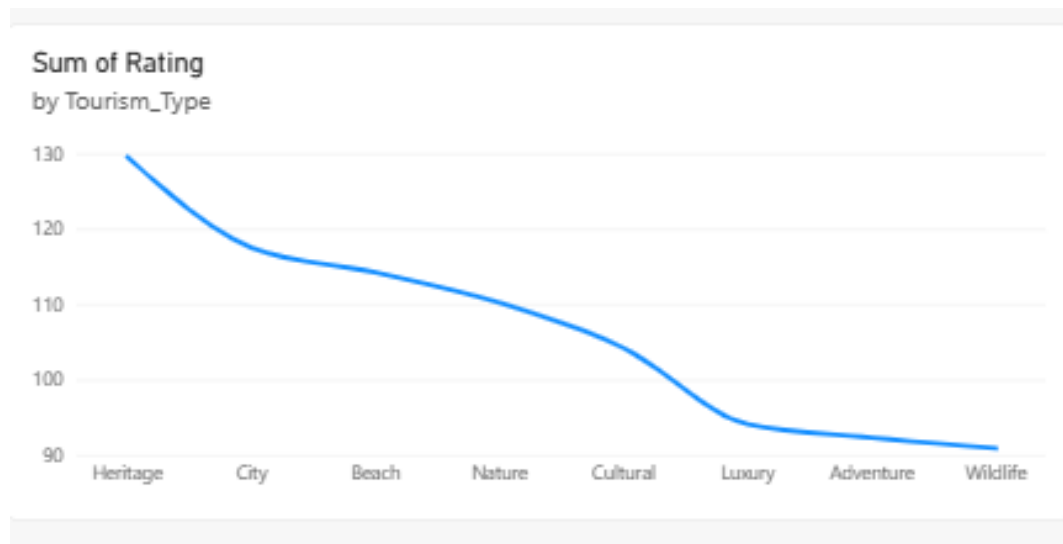
To clean, preprocess, and transform the raw **Tourism Dataset**, load the cleaned data model into **Power BI**, and build an interactive visual dashboard to analyze tourism destinations, tourism types, continents, estimated budgets, ratings, visitor statistics, accommodation types, and transportation modes.

Algorithm:

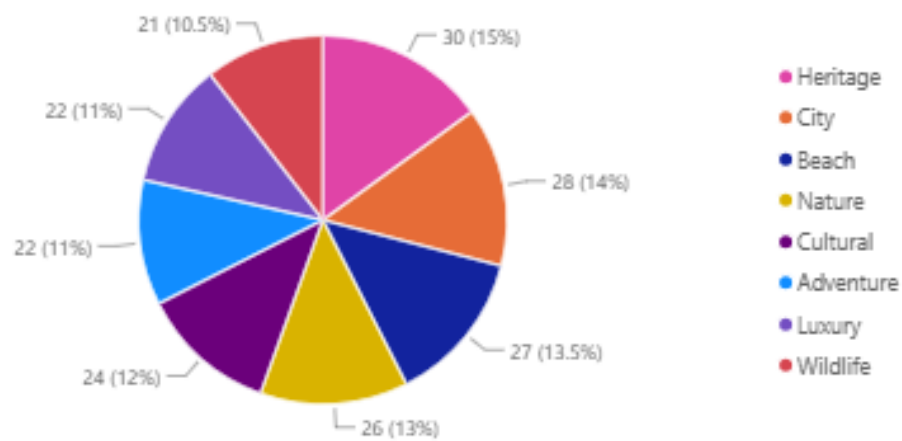
- **Start.**
- Open **Power BI Desktop**.
- Import the `tourism_dataset_200.csv` dataset using **Get Data** → **Text/CSV**.
- Load the dataset into the **Power Query Editor**.
- Inspect the dataset columns and verify the data.
- Identify and remove duplicate records, if any.
- Check for missing or null values in all columns.
- Handle missing values appropriately:
 - Replace missing numerical values using mean, median, or 0.
 - Replace missing categorical values using mode or "Unknown".
- Verify and correct the data types of all attributes.
- Set `Tourism_ID`, `Average_Duration_Days`, and `Annual_Visitors` as appropriate numeric data types.
- Set `Estimated_Budget_USD` and `Rating` as decimal numbers.
- Set categorical attributes such as `Destination`, `Country`, `Continent`, `Tourism_Type`, `Best_Season`, `Accommodation_Type`, and `Transportation` as text.
- Apply the transformations and load the cleaned dataset into Power BI using **Close & Apply**.
- Create DAX measures for:
 - Total Destinations
 - Average Estimated Budget
 - Average Rating
 - Total Annual Visitors
 - Average Duration
- Create a **Card visualization** to display the total number of tourism destinations.
- Create a **Bar Chart** to visualize tourism destinations by continent.
- Create a **Pie Chart** to display the distribution of tourism types.
- Create a **Column Chart** to compare the average estimated budget by continent.
- Create a **Bar Chart** to analyze the average rating by tourism type.
- Create a **Bar Chart** to visualize annual visitors by country.
- Create a **Scatter Plot** to analyze the relationship between estimated budget and rating.
- Create a **Pie Chart** to visualize the distribution of accommodation types.
- Create a **Donut Chart** to visualize different transportation modes.
- Add slicers for `Continent`, `Country`, `Tourism_Type`, `Best_Season`, and `Accommodation_Type`.

- Arrange all visualizations to create an interactive **Tourism Analysis Dashboard**.
- Analyze the dashboard and interpret the results.
- Save the Power BI report.
- **Stop.**

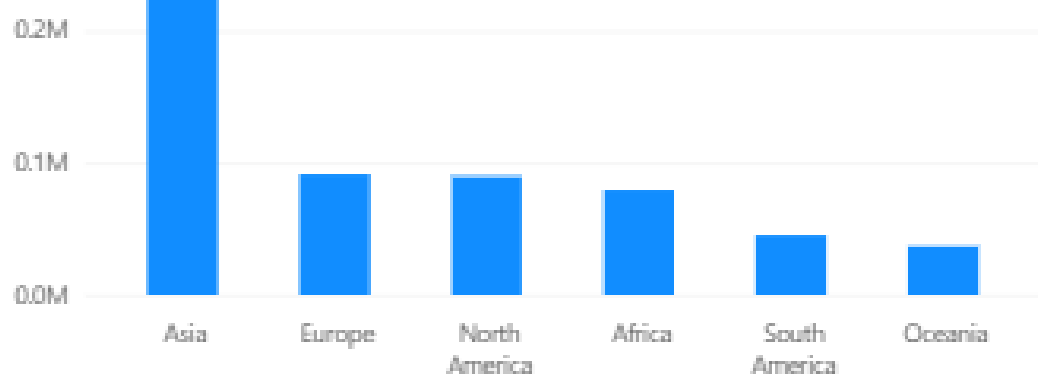
Output:

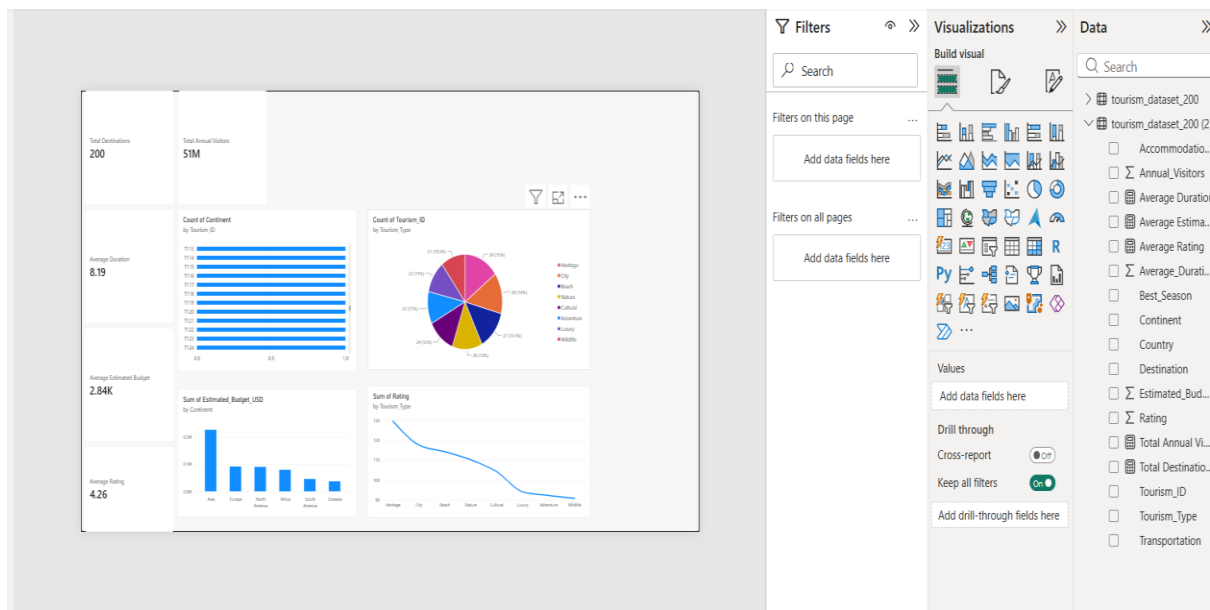


Count of Tourism_ID
by Tourism_Type



Sum of Estimated_Budget_USD
by Continent





Result:

Thus, the **Tourism Dataset** was successfully cleaned, preprocessed, transformed, and visualized using **Power BI**. An interactive dashboard was created to analyze tourism destinations, continents, tourism types, estimated budgets, ratings, annual visitors, accommodation types, and transportation modes.

Important: Since your Tourism Dataset contains **200 records and no missing values**, the missing-value handling steps may not actually modify any records, but you should still check for null values during the Power BI data-cleaning process.